# Separable Equations

**ANSWER KEY**

## Section 1: Separation Form and Integrals

**1** A first-order differential equation $\frac{dy}{dx} = f(x, y)$ is separable if it can be factored into:

**a.** $\frac{dy}{dx} = g(x)h(y)$ ✓

**b.** $\frac{dy}{dx} = g(x) + h(y)$

**c.** $\frac{dy}{dx} = g(x/y)$

**d.** $\frac{dy}{dx} = x^n + y^n$

A differential equation is separable if the right-hand side factors into a product of a function of $x$ alone and a function of $y$ alone: $g(x)h(y)$.

**2** Separate variables for $\frac{dy}{dx} = 3x^2y^2$.

**a.** $\frac{1}{y^2}dy = 3x^2dx$ ✓

**b.** $y^2dy = 3x^2dx$

**c.** $\frac{1}{3x^2}dy = y^2dx$

**d.** $dy = 3x^2y^2dx$

Divide both sides by $y^2$ and multiply by $dx$: $\frac{1}{y^2}dy = 3x^2dx$.

**3** Solve the general equation $\frac{1}{y^2}dy = 3x^2dx$.

**a.** $y = -\frac{1}{x^3 + C}$ ✓

**b.** $y = \frac{1}{x^3 + C}$

**c.** $y = -x^3 + C$

**d.** $y = \sqrt{x^3 + C}$

Integrate both sides: $-\frac{1}{y} = x^3 + C \Rightarrow y = -\frac{1}{x^3 + C}$.

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## Section 2: Initial Value Problems (IVPs)

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**4** Solve the IVP: $\frac{dy}{dx} = \frac{x}{y}$ with $y(0) = -3$.

**a.** $y = -\sqrt{x^2 + 9}$ ✓

**b.** $y = \sqrt{x^2 + 9}$

**c.** $y = -\sqrt{x^2 - 9}$

**d.** $y = -x - 3$

$ydy = xdx \Rightarrow \frac{1}{2}y^2 = \frac{1}{2}x^2 + C \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = -3$: $(-3)^2 = 0 + 2C \Rightarrow 2C = 9$. Thus $y^2 = x^2 + 9$. Since $y(0) = -3 < 0$, we select the negative branch $y = -\sqrt{x^2 + 9}$.

**5** Solve $\frac{dy}{dx} = 2y$ with $y(0) = 7$.

**a.** $y = 7e^{2x}$ ✓

**b.** $y = 7e^{-2x}$

**c.** $y = 2e^{7x}$

**d.** $y = 7 + 2x$

$\frac{1}{y}dy = 2dx \Rightarrow \ln|y| = 2x + C \Rightarrow y = Ae^{2x}$. Using $y(0) = 7$: $y = 7e^{2x}$.

**6** What is a singular solution of a differential equation?

**a.** **A solution that cannot be obtained from the general family for any value of the arbitrary constant** $C$ ✓

**b.** A solution that blows up to infinity everywhere

**c.** A solution where the derivative is undefined

**d.** A solution with no arbitrary constants

A singular solution (often an equilibrium solution where $h(y) = 0$) cannot be generated by tuning the parameter $C$ in the general solution formula.

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## Section 3: Exponential and Logistic Growth Models

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**7** What is the differential equation for logistic population growth with carrying capacity $K$ and growth rate $r$?

**a.** $\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$ ✓

**b.** $\frac{dP}{dt} = r(K - P)$

**c.** $\frac{dP}{dt} = rP^2$

**d.** $\frac{dP}{dt} = \frac{rP}{K}$

The logistic differential equation is $\frac{dP}{dt} = rP(1 - P/K)$, where growth slows down as population $P$ approaches carrying capacity $K$.

**8** At what population size does the logistic growth rate $\frac{dP}{dt}$ reach its maximum?

**a.** $P = \frac{K}{2}$ ✓

**b.** $P = K$

**c.** $P = 0$

**d.** $P = \frac{K}{4}$

The quadratic $f(P) = rP - \frac{r}{K}P^2$ reaches its maximum at vertex $P = -\frac{r}{2(-r/K)} = \frac{K}{2}$.

**9** What is $\lim_{t \to \infty} P(t)$ in a logistic model if $P(0) > 0$?

**a.** $K$ ✓

**b.** $\infty$

**c.** $0$

**d.** $rK$

For any positive initial population, the population asymptotically approaches the stable equilibrium carrying capacity $K$ as $t \to \infty$.

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## Section 4: Newton's Law of Cooling

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**10** Newton's Law of Cooling states that the rate of change of temperature $T$ is proportional to the difference between $T$ and the ambient temperature $T_m$:

**a.** $\frac{dT}{dt} = -k(T - T_m)$ **with** $k > 0$ ✓

**b.** $\frac{dT}{dt} = -kT$

**c.** $\frac{dT}{dt} = k(T + T_m)$

**d.** $\frac{dT}{dt} = -k(T - T_m)^2$

The rate of cooling is proportional to the temperature gradient with the surroundings: $\frac{dT}{dt} = -k(T - T_m)$.

**11** A metal bar at $100^\circ\text{C}$ is placed in a room at $20^\circ\text{C}$. If $T(t) = 20 + 80e^{-kt}$ and cools to $60^\circ\text{C}$ after $10$ minutes, find $e^{-10k}$.

**a.** $\frac{1}{2}$ ✓

**b.** $\frac{3}{4}$

**c.** $\frac{1}{4}$

**d.** $\frac{2}{3}$

$60 = 20 + 80e^{-10k} \Rightarrow 40 = 80e^{-10k} \Rightarrow e^{-10k} = \frac{40}{80} = \frac{1}{2}$.

**12** Using $e^{-10k} = \frac{1}{2}$ from above, what is the temperature after $20$ minutes?

**a.** $40^\circ\text{C}$ ✓

**b.** $30^\circ\text{C}$

**c.** $50^\circ\text{C}$

**d.** $25^\circ\text{C}$

After 20 minutes (two 10-minute half-cooling intervals): $e^{-20k} = (e^{-10k})^2 = (1/2)^2 = 1/4$. Temperature $T(20) = 20 + 80(1/4) = 20 + 20 = 40^\circ\text{C}$.

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